



Faculty of Engineering Sciences and Technology
CURRICULUM FOR BS (ARTIFICIAL INTELLIGENCE) PROGRAM
(04 Year Degree Program)

Semester	Course(s) Code (New)	Course(s) Title	Credit Hours	Prerequisite
Semester I	CSC101	Applied Physics	3 + 1	---
	CSC111	Introduction to Information and Communication Technology	3 + 1	---
	CSC102	Calculus and Analytical Geometry	3 + 0	---
	HUM111	Functional English	3 + 0	---
	HUM122	Islamiat and Pakistan Studies	3 + 0	---
		Total Semester Credit Hours	(15 + 2)	
Semester II	CSC103	Linear Algebra and Differential Equations	3 + 0	CSC102
	HUM231	Communication Skills	3 + 0	HUM111
	CSC131	Programming Fundamentals	3 + 1	CSC111
	CSC104	Probability and Statistics	3 + 0	CSC104
	HUM112	Personal Development	3 + 0	---
		Total Semester Credit Hours	(15 + 1)	
Semester III	CSC221	Operating Systems	3 + 1	CSC131
	CSC231	Object Oriented Programming	3 + 1	CSC131
	CSC201	Multivariate Calculus	3 + 0	CSC102
	HUM233	Philosophy and Critical Thinking	3 + 0	---
	SSC231	World History	3 + 0	HUM111
		Total Semester Credit Hours	(15 + 2)	
Semester IV	CSC232	Data Structures & Algorithms	3 + 1	CSC231
	CSC233	Database Management Systems	3 + 1	CSC221 CSC231
	CSC241	Discrete Structures	3 + 0	CSC102
	AIN271	Applied Statistics	3 + 0	CSC102
	CSC261	Data Communication and Computer Networks	3 + 1	CSC221
	HUM113	Sociology	3 + 0	---

Semester	Course(s) Code (New)	Course(s) Title	Credit Hours	Prerequisite
		Total Semester Credit Hours	(18 + 3)	
Semester V	AIN371	Artificial Intelligence	3 + 1	AIN271, CSC232
	CSC331	Design and Analysis of Algorithms	3 + 0	CSC232
	AIN372	Continuous and Discrete Optimization	3 + 1	CSC201, CSC231
	HUM241	World Literature	3 + 0	HUM111
	CSC351	Software Engineering	3 + 0	CSC231
		Total Semester Credit Hours	(15 + 2)	
Semester VI	AIN373	Machine Learning	3 + 1	AIN371, AIN372
	AIN374	Artificial Intelligence for Robotics	3 + 1	AIN371
	HUM121	Academic Writing	3 + 0	HUM111
	AIN375	Data Visualization	3 + 1	CSC231, AIN271
	CSC332	Parallel and Distributed Computing	3 + 1	CSC221, CSC231, CSC261
		Total Semester Credit Hours	(15 + 4)	
Semester VII	CSC441	Final Year Project – I	0 + 3	CSC351
	AIN471	Logic & Automated Reasoning	3 + 0	AIN371
		Elective I	3 + 0	
	AIN472	Deep Learning	3 + 1	AIN373
		Elective II	3 + 0	
		Total Semester Credit Hours	(12+ 4)	
Semester VIII		Elective III	3 + 0	
	CSC473	Computer Vision	3 + 1	AIN472
	CSC442	Final Year Project – II	0 + 3	CSC441
	CSC474	Natural Language Processing	3 + 1	AIN472
	HUM232	Ethics & Social Responsibility	3 + 0	---
		Total Semester Credit Hours	(12 + 5)	
Program Total Credit Hour(s): 140				

List of Elective Courses for BS (Artificial Intelligence) Program

Course Code	Course(s) Title	Credit Hours
AIN477	Security Analytics	3 + 0
AIN478	Reinforcement Learning	3 + 0
AIN479	Image Processing & Analysis	3 + 0
AIN480	Probabilistic Graphical Models	3 + 0
AIN481	High-Performance Computing	3 + 0
AIN482	Big Data Analytics	3 + 0
AIN483	Social Media Mining	3 + 0
AIN484	Metaheuristic Optimization	3 + 0
AIN485	Security & Privacy	3 + 0

Course Area Category:

- 0-Maths & Science.
- 1-Information Technology.
- 2-Systems.
- 3-Programming & Databases.
- 4-Computer Science.
- 5-Software Engineering.
- 6-Networks & Communication.
- 7- Artificial Intelligence.
- 8-Graphics & Animation